

As the point w goes to infinity in $\mathbb{C}$ (in the sense that $|w| \rightarrow \infty$) the corresponding point W on $\mathbb{S}$ comes arbitrarily close to $\mathcal{N}$. This simple observation makes $\mathcal{N}$ a natural candidate for the so-called “point at infinity.” Identifying infinity with the point $\mathcal{N}$ on $\mathbb{S}$, we see that the extended complex plane can be visualized as the full two-dimensional sphere $\mathbb{S}$; this is the **Riemann sphere**. Since this construction takes the unbounded set $\mathbb{C}$ into the compact set $\mathbb{S}$ by adding one point, the Riemann sphere is sometimes called the **one-point compactification** of $\mathbb{C}$.

An important consequence of this interpretation is the following: although the point at infinity required special attention when considered separately from $\mathbb{C}$, it now finds itself on equal footing with all other points on $\mathbb{S}$. In particular, a meromorphic function on the extended complex plane can be thought of as a map from $\mathbb{S}$ to itself, where the image of a pole is now a tractable point on $\mathbb{S}$, namely $\mathcal{N}$. For these reasons (and others) the Riemann sphere provides good geometrical insight into the structure of $\mathbb{C}$ as well as the theory of meromorphic functions.

4 The argument principle and applications

We anticipate our discussion of the logarithm (in Section 6) with a few comments. In general, the function $\log f(z)$ is “multiple-valued” because it cannot be defined unambiguously on the set where $f(z) \neq 0$. However it is to be defined, it must equal $\log |f(z)| + i \arg f(z)$, where $\log |f(z)|$ is the usual real-variable logarithm of the positive quantity $|f(z)|$ (and hence is defined unambiguously), while $\arg f(z)$ is some determination of the argument (up to an additive integral multiple of 2π). Note that in any case, the derivative of $\log f(z)$ is $f'(z)/f(z)$ which is single-valued, and the integral

$$\int_{\gamma} \frac{f'(z)}{f(z)} dz$$

can be interpreted as the change in the argument of f as z traverses the curve γ . Moreover, assuming the curve is closed, this change of argument is determined entirely by the zeros and poles of f inside γ . We now formulate this fact as a precise theorem.

We begin with the observation that while the additivity formula

$$\log(f_1 f_2) = \log f_1 + \log f_2$$

fails in general (as we shall see below), the additivity can be restored to the corresponding derivatives. This is confirmed by the following

observation:

$$\frac{(f_1 f_2)'}{f_1 f_2} = \frac{f_1' f_2 + f_1 f_2'}{f_1 f_2} = \frac{f_1'}{f_1} + \frac{f_2'}{f_2},$$

which generalizes to

$$\frac{\left(\prod_{k=1}^N f_k\right)'}{\prod_{k=1}^N f_k} = \sum_{k=1}^N \frac{f_k'}{f_k}.$$

We apply this formula as follows. If f is holomorphic and has a zero of order n at z_0 , we can write

$$f(z) = (z - z_0)^n g(z),$$

where g is holomorphic and nowhere vanishing in a neighborhood of z_0 , and therefore

$$\frac{f'(z)}{f(z)} = \frac{n}{z - z_0} + G(z)$$

where $G(z) = g'(z)/g(z)$. The conclusion is that if f has a zero of order n at z_0 , then f'/f has a simple pole with residue n at z_0 . Observe that a similar fact also holds if f has a pole of order n at z_0 , that is, if $f(z) = (z - z_0)^{-n} h(z)$. Then

$$\frac{f'(z)}{f(z)} = \frac{-n}{z - z_0} + H(z).$$

Therefore, if f is meromorphic, the function f'/f will have simple poles at the zeros and poles of f , and the residue is simply the order of the zero of f or the negative of the order of the pole of f . As a result, an application of the residue formula gives the following theorem.

Theorem 4.1 (Argument principle) *Suppose f is meromorphic in an open set containing a circle C and its interior. If f has no poles and never vanishes on C , then*

$$\frac{1}{2\pi i} \int_C \frac{f'(z)}{f(z)} dz = (\text{number of zeros of } f \text{ inside } C) \text{ minus} \\ (\text{number of poles of } f \text{ inside } C),$$

where the zeros and poles are counted with their multiplicities.

Corollary 4.2 *The above theorem holds for toy contours.*

As an application of the argument principle, we shall prove three theorems of interest in the general theory. The first, Rouché's theorem, is in some sense a continuity statement. It says that a holomorphic function can be perturbed slightly without changing the number of its zeros. Then, we prove the open mapping theorem, which states that holomorphic functions map open sets to open sets, an important property that again shows the special nature of holomorphic functions. Finally, the maximum modulus principle is reminiscent of (and in fact implies) the same property for harmonic functions: a non-constant holomorphic function on an open set Ω cannot attain its maximum in the interior of Ω .

Theorem 4.3 (Rouché's theorem) *Suppose that f and g are holomorphic in an open set containing a circle C and its interior. If*

$$|f(z)| > |g(z)| \quad \text{for all } z \in C,$$

then f and $f + g$ have the same number of zeros inside the circle C .

Proof. For $t \in [0, 1]$ define

$$f_t(z) = f(z) + tg(z)$$

so that $f_0 = f$ and $f_1 = f + g$. Let n_t denote the number of zeros of f_t inside the circle counted with multiplicities, so that in particular, n_t is an integer. The condition $|f(z)| > |g(z)|$ for $z \in C$ clearly implies that f_t has no zeros on the circle, and the argument principle implies

$$n_t = \frac{1}{2\pi i} \int_C \frac{f'_t(z)}{f_t(z)} dz.$$

To prove that n_t is constant, it suffices to show that it is a continuous function of t . Then we could argue that if n_t were not constant, the intermediate value theorem would guarantee the existence of some $t_0 \in [0, 1]$ with n_{t_0} not integral, contradicting the fact that $n_t \in \mathbb{Z}$ for all t .

To prove the continuity of n_t , we observe that $f'_t(z)/f_t(z)$ is jointly continuous for $t \in [0, 1]$ and $z \in C$. This joint continuity follows from the fact that it holds for both the numerator and denominator, and our assumptions guarantee that $f_t(z)$ does not vanish on C . Hence n_t is integer-valued and continuous, and it must be constant. We conclude that $n_0 = n_1$, which is Rouché's theorem.

We now come to an important geometric property of holomorphic functions that arises when we consider them as mappings (that is, mapping regions in the complex plane to the complex plane).

A mapping is said to be **open** if it maps open sets to open sets.

Theorem 4.4 (Open mapping theorem) *If f is holomorphic and non-constant in a region Ω , then f is open.*

Proof. Let w_0 belong to the image of f , say $w_0 = f(z_0)$. We must prove that all points w near w_0 also belong to the image of f .

Define $g(z) = f(z) - w$ and write

$$\begin{aligned} g(z) &= (f(z) - w_0) + (w_0 - w) \\ &= F(z) + G(z). \end{aligned}$$

Now choose $\delta > 0$ such that the disc $|z - z_0| \leq \delta$ is contained in Ω and $f(z) \neq w_0$ on the circle $|z - z_0| = \delta$. We then select $\epsilon > 0$ so that we have $|f(z) - w_0| \geq \epsilon$ on the circle $|z - z_0| = \delta$. Now if $|w - w_0| < \epsilon$ we have $|F(z)| > |G(z)|$ on the circle $|z - z_0| = \delta$, and by Rouché's theorem we conclude that $g = F + G$ has a zero inside the circle since F has one.

The next result pertains to the size of a holomorphic function. We shall refer to the **maximum** of a holomorphic function f in an open set Ω as the maximum of its absolute value $|f|$ in Ω .

Theorem 4.5 (Maximum modulus principle) *If f is a non-constant holomorphic function in a region Ω , then f cannot attain a maximum in Ω .*

Proof. Suppose that f did attain a maximum at z_0 . Since f is holomorphic it is an open mapping, and therefore, if $D \subset \Omega$ is a small disc centered at z_0 , its image $f(D)$ is open and contains $f(z_0)$. This proves that there are points in $z \in D$ such that $|f(z)| > |f(z_0)|$, a contradiction.

Corollary 4.6 *Suppose that Ω is a region with compact closure $\overline{\Omega}$. If f is holomorphic on Ω and continuous on $\overline{\Omega}$ then*

$$\sup_{z \in \Omega} |f(z)| \leq \sup_{z \in \overline{\Omega} - \Omega} |f(z)|.$$

In fact, since $f(z)$ is continuous on the compact set $\overline{\Omega}$, then $|f(z)|$ attains its maximum in $\overline{\Omega}$; but this cannot be in Ω if f is non-constant. If f is constant, the conclusion is trivial.

Remark. The hypothesis that $\overline{\Omega}$ is compact (that is, bounded) is essential for the conclusion. We give an example related to considerations that we will take up in Chapter 4. Let Ω be the open first quadrant, bounded by the positive half-line $x \geq 0$ and the corresponding imaginary line $y \geq 0$. Consider $F(z) = e^{-iz^2}$. Then F is entire and clearly

continuous on $\overline{\Omega}$. Moreover $|F(z)| = 1$ on the two boundary lines $z = x$ and $z = iy$. However, $F(z)$ is unbounded in Ω , since for example, we have $F(z) = e^{r^2}$ if $z = r\sqrt{i} = re^{i\pi/4}$.

5 Homotopies and simply connected domains

The key to the general form of Cauchy's theorem, as well as the analysis of multiple-valued functions, is to understand in what regions we can define the primitive of a given holomorphic function. Note the relevance to the study of the logarithm, which arises as a primitive of $1/z$. The question is not just a local one, but is also global in nature. Its elucidation requires the notion of homotopy, and the resulting idea of simple-connectivity.

Let γ_0 and γ_1 be two curves in an open set Ω with common end-points. So if $\gamma_0(t)$ and $\gamma_1(t)$ are two parametrizations defined on $[a, b]$, we have

$$\gamma_0(a) = \gamma_1(a) = \alpha \quad \text{and} \quad \gamma_0(b) = \gamma_1(b) = \beta.$$

These two curves are said to be **homotopic in Ω** if for each $0 \leq s \leq 1$ there exists a curve $\gamma_s \subset \Omega$, parametrized by $\gamma_s(t)$ defined on $[a, b]$, such that for every s

$$\gamma_s(a) = \alpha \quad \text{and} \quad \gamma_s(b) = \beta,$$

and for all $t \in [a, b]$

$$\gamma_s(t)|_{s=0} = \gamma_0(t) \quad \text{and} \quad \gamma_s(t)|_{s=1} = \gamma_1(t).$$

Moreover, $\gamma_s(t)$ should be jointly continuous in $s \in [0, 1]$ and $t \in [a, b]$.

Loosely speaking, two curves are homotopic if one curve can be deformed into the other by a continuous transformation without ever leaving Ω (Figure 6).

Theorem 5.1 *If f is holomorphic in Ω , then*

$$\int_{\gamma_0} f(z) dz = \int_{\gamma_1} f(z) dz$$

whenever the two curves γ_0 and γ_1 are homotopic in Ω .

Proof. The key to the proof lies in showing that if two curves are close to each other and have the same end-points, then the integrals over these curves are equal. Recall that by definition, the function $F(s, t) = \gamma_s(t)$ is continuous on $[0, 1] \times [a, b]$. In particular, since the image of F , which we